ID: db876fd5

Songbirds learn to respond to and imitate their species' songs from an early age. With each generation, small differences are introduced that result in distinct variations—called dialects—among geographically isolated populations of the same species. A research study examined whether twelve-day-old *Ficedula hypoleuca* (pied flycatcher) nestlings prefer local dialects over the unfamiliar dialects of nonlocal *F. hypoleuca* populations: the more begging calls the nestlings made in response to a song, the stronger their preference. The researchers found that nestlings produced more begging calls in response to their own dialect than to nonlocal dialects. Since song preference plays a role in songbird mate selection, the finding suggests that _____

Which choice most logically completes the text?

A. *F. hypoleuca* nestlings' preference for their own dialect likely disappears as they mature to promote socialization between different *F. hypoleuca* populations.

B. *F. hypoleuca* nestlings who show an early preference for their own dialect are likely to receive more food from their caretakers than nestlings who show no preferences among any *F. hypoleuca* dialects.

C. *F. hypoleuca* nestlings' preference for their own dialect likely drives them when they mature to reproduce with other *F. hypoleuca* from local rather than nonlocal populations.

D. *F. hypoleuca* nestlings show a preference for both local *F. hypoleuca* dialects and the songs of other local songbirds over the songs of nonlocal birds of any species.

ID: 575e67df

By running computer simulations of the development of our solar system, André Izidoro, Rajdeep Dasgupta, and colleagues concluded that the Sun may have been surrounded by three giant dust rings before the planets started to form. The researchers suggest that the materials in the innermost ring became the four planets closest to the Sun, the materials in the middle ring produced the rest of the planets, and the materials in the outermost ring created the asteroids and other small bodies in the region beyond Neptune. In one simulation, the researchers delayed the initial formation of the middle ring, causing oversized super-Earths to begin developing from the innermost ring. The researchers therefore hypothesize that _____

Which choice most logically completes the text?

A. the middle ring formed earlier in the solar system's development than the initial simulations suggested.

B. the timing of the initial formation of the middle ring played an important role in determining the eventual size of Earth.

C. if the formation of the outermost ring had occurred earlier in a simulation, all the planets would have become super-Earths.

D. the innermost ring actually formed into all the planets in our solar system, not just the four closest to the Sun.

ID: a59245a1

The painter María Izquierdo played an important role in the development of twentieth-century Mexican art, but her work has never been well-known in the United States except among art historians. One reason for Izquierdo's relative obscurity is the enormous popularity of some of her peers. In particular, the painters Frida Kahlo and Diego Rivera have so captivated the interest of US audiences that Izquierdo and other Mexican artists from the period often get overlooked, despite the high quality of their work.

Which choice best states the main idea of the text?

A. Izquierdo's work is not as well-known in the United States as it should be because Kahlo and Rivera draw so much of the public's attention.

B. During Izquierdo's lifetime, her paintings were displayed in galleries in the United States much more frequently than paintings by Kahlo and Rivera were.

C. Izquierdo painted some of the same subjects that Kahlo and Rivera painted but used different techniques than they used.

D. Few of Izquierdo's works are in galleries today because she produced only a small number of paintings.

ID: c538954d

Sense and Sensibility is an 1811 novel by Jane Austen. In the novel, Austen describes Marianne Dashwood's ability to persuade others of the rightness of her artistic judgments, as is evident when Marianne visits with John Willoughby, a potential suitor: _____

Which quotation from *Sense and Sensibility* most effectively illustrates the claim?

A. "Above all, when she heard him declare, that of music and dancing he was passionately fond, she gave him such a look of approbation as secured the largest share of his discourse to herself for the rest of his stay."

B. "Their taste was strikingly alike. The same books, the same passages were idolized by each—or if any difference appeared, any objection arose, it lasted no longer than till the force of her arguments and the brightness of her eyes could be displayed."

C. "It was only necessary to mention any favourite amusement to engage her to talk. She could not be silent when such points were introduced, and she had neither shyness nor reserve in their discussion."

D. "They speedily discovered that their enjoyment of dancing and music was mutual, and that it arose from a general conformity of judgment in all that related to either. Encouraged by this to a further examination of his opinions, she proceeded to question him on the subject of books."

ID: 25290c8d

"On Virtue" is a 1766 poem by Phillis Wheatley. Wheatley addresses the poem directly to the quality of virtue, imploring it to assist her in reaching a future goal: _____

Which quotation from "On Virtue" most effectively illustrates the claim?

A. "Attend me, *Virtue*, thro' my youthful years! / O leave me not to the false joys of time! / But guide my steps to endless life and bliss."

B. "I cease to wonder, and no more attempt / Thine height t'explore, or fathom thy profound."

C. "O thou bright jewel in my aim I strive / To comprehend thee. Thine own words declare / Wisdom is higher than a fool can reach."

D. "But, O my soul, sink not into despair, / *Virtue* is near thee, and with gentle hand / Would now embrace thee, hovers o'er thine head."

ID: 11a9f635

Paleontologists searching for signs of ancient life have found many fossilized specimens of prehistoric human ancestors, including several from the Pleistocene era discovered in a geological formation in the Minatogawa quarry in Japan. However, to study the emergence of the earliest multicellular organisms to appear on Earth, researchers must turn elsewhere, such as to the Ediacaran geological formation at Mistaken Point in Canada. A UNESCO World Heritage Site, the 146-hectare reserve contains more than 10,000 fossils that together document a critical moment in evolutionary history.

What does the text indicate about the geological formation at Mistaken Point?

A. It holds a greater number of fossils but from a smaller variety of species than the formation in the Minatogawa quarry does.

B. It has provided evidence that the earliest human species may have emerged before the Pleistocene era.

C. It is widely considered by paleontologists to be the most valuable source of information about prehistoric life forms.

D. It contains specimens from an older time period than those found in the formation in the Minatogawa quarry.

ID: 7cbb9764

Accomplished printmaker and sculptor Elizabeth Catlett (1915–2012) used her art to explore the Black experience in the United States. In a paper for an art history class, a student claims that Catlett had a particular talent for unifying various artistic traditions and styles in her work.

Which quotation from a scholar describing Catlett's work would best support the student's claim?

A. "In *Mother and Child*, a sculpture of two Black figures, Catlett used an ancient Indigenous sculpting technique and combined the visual aesthetic of modern Mexican muralists with that of German artist Kathe Kollwitz."

B. "In her collage *New Generation*, Catlett overlaid fabric onto the canvas to represent the clothing of a father and his toddler, positioned to evoke classic images of a mother and child."

C. "Created in 1968, Catlett's sculpture *Black Unity*, a stylized fist sculpted from mahogany and measuring two feet across, remains an important piece and has received renewed and well-deserved attention in recent years."

D. "One series of Catlett's prints, made by the artist using the linoleum cut method, depicts several notable African American women, including Harriet Tubman and Sojourner Truth."

ID: dd349efc

Participants' Evaluation of the Likelihood That Robots Can Work Effectively in Different Occupations

Occupation	Somewhat or very unlikely (%)	Neutral (%)	Somewhat or very likely (%)
television news anchor	24	9	67
teacher	37	16	47
firefighter	62	9	30
surgeon	74	9	16
tour guide	10	8	82

Rows in table may not add up to 100 due to rounding.

Georgia Tech roboticists De'Aira Bryant and Ayanna Howard, along with ethicist Jason Borenstein, were interested in people's perceptions of robots' competence. They recruited participants and asked them how likely they think it is that a robot could do the work required in various occupations. Participants' evaluations varied widely depending on which occupation was being considered; for example, _____

Which choice most effectively uses data from the table to complete the example?

A. 47% of participants believe that it is somewhat or very likely that a robot could work effectively as a teacher, but 37% of respondents believe that it is somewhat or very unlikely that a robot could do so.

B. 9% of participants were neutral about whether a robot could work effectively as a television news anchor, which is the same percent of participants who were neutral when asked about a robot working as a surgeon.

C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.

D. 62% of participants believe that it is somewhat or very unlikely that a robot could work effectively as a firefighter.

ID: 8fbed1cb

When the Vinland Map, a map of the world purported to date to the mid-1400s, surfaced in 1957, some scholars believed it demonstrated that European knowledge of the eastern coast of present-day North America predated Christopher Columbus's 1492 arrival. In 2021, a team including conservators Marie-France Lemay and Paula Zyats and materials scientist Anikó Bezur performed an extensive analysis of the map and the ink used. They found that the ink contains titanium dioxide, a compound that was first introduced in ink manufacturing in the early 1900s. Therefore, the team concluded that _____

Which choice most logically completes the text?

A. mid-1400s Europeans could not have known about the eastern coast of present-day North America.

B. the Vinland Map could not have been drawn by mid-1400s mapmakers.

C. mapmakers must have used titanium compounds in their ink in the 1400s.

D. there isn't enough information to determine when the ink was created.

ID: dc3ea63e

To dye wool, Navajo (Diné) weaver Lillie Taylor uses plants and vegetables from Arizona, where she lives. For example, she achieved the deep reds and browns featured in her 2003 rug *In the Path of the Four Seasons* by using Arizona dock roots, drying and grinding them before mixing the powder with water to create a dye bath. To intensify the appearance of certain colors, Taylor also sometimes mixes in clay obtained from nearby soil.

Which choice best states the main idea of the text?

A. Reds and browns are not commonly featured in most of Taylor's rugs.

B. Taylor draws on local resources in the approach she uses to dye wool.

C. Taylor finds it difficult to locate Arizona dock root in the desert.

D. *In the Path of the Four Seasons* is widely acclaimed for its many colors and innovative weaving techniques.

ID: 5432d1de

It's common for jazz musicians and fans to refer to certain songs as having "swing," indicating that the songs provoke a strong feeling, like the impulse to tap one's foot or dance. The exact acoustic properties that give a song swing, however, have long been thought to be undefinable. To investigate swing, a team led by physicist Corentin Nelias delayed the downbeats and synchronized the offbeats in jazz piano solos and asked jazz musicians to compare the intensity of swing in each modified piece with the intensity of swing in the original piece. They found that participants were more than seven times likelier to characterize the modified songs as having swing than to characterize the original versions as having swing, suggesting that _____

Which choice most logically completes the text?

A. synchronized offbeats tend to give a song swing regardless of whether downbeats are delayed.

B. the acoustic properties that give a song swing are not easy for jazz musicians to manipulate.

C. jazz songs that feature the piano are more likely to have swing than are jazz songs that do not feature the piano.

D. the timing of downbeats and offbeats may play a crucial role in giving a song swing.

ID: a13541c0

Sandra Cisneros's 1984 novella *The House on Mango Street* made a lasting impact on US literature. Its depiction of Mexican American culture inspired later authors to examine their own heritage within their fictional works. Also influential was the book's portrayal of the main character, Esperanza, during a pivotal year of her youth. This insightful depiction of a preteen girl encouraged authors who, like Cisneros herself, are Latina to use fictional works to examine experiences from their own youth.

Which statement, if true, would most strongly support the claim in the underlined sentence?

A. In interviews, a number of Latina authors say that *The House on Mango Street* inspired them to write about their own adolescence in their novels.

B. In published writings, several prominent authors who are not Latina say that reading *The House on Mango Street* influenced their approach to writing fiction.

C. The *House on Mango Street* has sold over six million copies and is one of the most commonly read books among high school and university students in the US.

D. Since 1984, new novels about young Latina characters by Latina authors have often been compared to *The House on Mango Street*.

ID: 4e9afd7a

The Indus River valley civilization flourished in South Asia from 3300 BCE to 1300 BCE. Many examples of the civilization's writing system exist, but researchers haven't yet deciphered it or identified which ancient language it represents. Nevertheless, archaeologists have found historical artifacts, such as clay figures and jewelry, that provide information about the civilization's customs and how its communities were organized. The archaeologists' findings therefore suggest that _____

Which choice most logically completes the text?

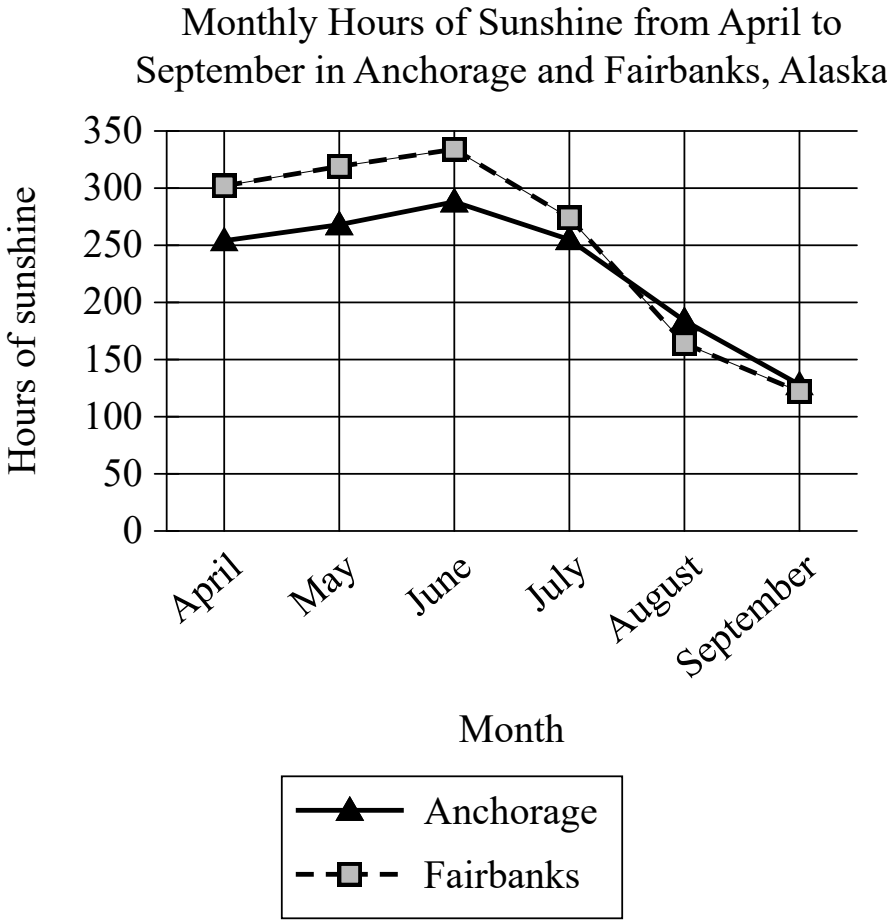
A. investigating an ancient civilization is easier without knowledge of the civilization's language.

B. knowing an ancient civilization's language isn't necessary in order to learn details about the civilization.

C. archaeological research should focus on finding additional artifacts rather than deciphering ancient languages.

D. examining the civilization's historical artifacts has resolved the debate about this civilization's language.

ID: 8a668840



A student is researching monthly hours of sunshine in different cities in Alaska. When comparing trends in Anchorage and Fairbanks, the student concludes that the two cities show a similar pattern in the monthly hours of sunshine from April to September.

Which choice best describes data from the graph that support the student's conclusion?

A. The monthly hours of sunshine in both Anchorage and Fairbanks hold steady in June and July before beginning to decline in August.

B. The monthly hours of sunshine in both Anchorage and Fairbanks increase from April to June and then decrease from June to September.

C. Anchorage and Fairbanks both have less than 200 monthly hours of sunshine from April to September.

D. Anchorage and Fairbanks both have more than 300 monthly hours of sunshine from April to June and less than 200 hours from July to September.

ID: dd72993d

Rivers rich in sediment appear yellow, while increases in red algae make rivers appear red. To track things like the sediment or algae content of large US rivers, John R. Gardner and colleagues used satellite data to determine the dominant visible wavelengths of light measured for various segments of these rivers. The researchers classified wavelengths of 495 nanometers (nm) and below as red, wavelengths between 495 and 560 nm as blue, and wavelengths of 560 nm and above as yellow. The researchers concluded that for the Missouri River, segments flowing into lakes tend to carry more sediment than those flowing out of lakes.

Which finding, if true, would most directly support the researchers' conclusion?

A. The segments of the Missouri River that had higher levels of chlorophyll-a, which contributes to the green color of photosynthetic organisms, have dominant wavelengths of light between 490 and 560 nm.

B. In lakes through which segments of the Missouri River pass, the dominant wavelength of light tended to be above 560 nm near the lakes' shores and below 560 nm in the lakes' centers.

C. The majority of the segments of the Missouri River were found to have dominant wavelengths of light significantly higher than 560 nm.

D. Segments of the Missouri River flowing into lakes typically had dominant wavelengths of light above 560 nm, while segments flowing out of lakes typically had dominant wavelengths below 560 nm.

ID: 3190835d

Some businesses believe that when employees are interrupted while doing their work, they experience a decrease in energy and productivity. However, a team led by Harshad Puranik, who studies management, has found that interruptions by colleagues can have a social component that increases employees' sense of belonging, resulting in greater job satisfaction that benefits employees and employers. Therefore, businesses should recognize that _____

Which choice most logically completes the text?

A. the interpersonal benefits of some interruptions in the workplace may offset the perceived negative effects.

B. in order to maximize productivity, employers should be willing to interrupt employees frequently throughout the day.

C. most employees avoid interrupting colleagues because they don't appreciate being interrupted themselves.

D. in order to cultivate an ideal workplace environment, interruptions of work should be discouraged.

ID: 9aa5efc4

Born in 1891 to a Quechua-speaking family in the Andes Mountains of Peru, Martín Chambi is today considered to be one of the most renowned figures of Latin American photography. In a paper for an art history class, a student claims that Chambi's photographs have considerable ethnographic value—in his work, Chambi was able to capture diverse elements of Peruvian society, representing his subjects with both dignity and authenticity.

Which finding, if true, would most directly support the student's claim?

A. Chambi took many commissioned portraits of wealthy Peruvians, but he also produced hundreds of images carefully documenting the peoples, sites, and customs of Indigenous communities of the Andes.

B. Chambi's photographs demonstrate a high level of technical skill, as seen in his strategic use of illumination to create dramatic light and shadow contrasts.

C. During his lifetime, Chambi was known and celebrated both within and outside his native Peru, as his work was published in places like Argentina, Spain, and Mexico.

D. Some of the peoples and places Chambi photographed had long been popular subjects for Peruvian photographers.

ID: b1fab3e1

Violins made by Antonio Stradivari and other craftspeople in the sixteenth to eighteenth centuries in Cremona, Italy, produce a sound that is considered superior to that of modern stringed instruments. Some experts have claimed that the type of wood used to create Cremonese violins is responsible for their prized sound, but modern and Cremonese violins are made of the same kinds of wood: maple and spruce. New analysis, however, has revealed unique indications that the wood in the older violins was chemically treated by the makers, leading researchers to suggest that _____

Which choice most logically completes the text?

A. Cremonese violins probably were not considered superior to other instruments at the time they were made.

B. the sound quality of Cremonese violins results in part from a method the craftspeople used to alter the wood.

C. if modern violins were made of a wood other than maple or spruce, they likely would sound as good as Cremonese violins.

D. the current process of making violins is the same process that was used centuries ago by Cremonese craftspeople.

ID: 55688b3c

Dutch painters in the sixteenth and seventeenth centuries often showed tables filled with large wheels of cheese or carved shards of butter. Some art historians, noting that dairy products were a major component of the Dutch diet, interpret these depictions as reflections of everyday Dutch eating habits. However, a group of researchers recently reviewed hundreds of food-related paintings and found that lemons—which could only be acquired in the Netherlands at great cost, since they had to be imported from warmer climates—feature in Dutch paintings of the period more than three times as frequently as dairy products do, thereby casting doubt on the idea that _____

Which choice most logically completes the text?

A. dairy products were a more significant component of the Dutch diet of the period than lemons were.

B. food was a more popular subject among Dutch painters than it was among painters from other countries at the time.

C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.

D. Dutch painters of the period may have depicted foods for symbolic reasons rather than to show what Dutch people typically ate.

ID: 350e2336

The Haitian Declaration of Independence was issued in 1804, bringing to an end the revolution against colonial France that began in 1791. Written in French, which was not the first language of most Haitians but which was used throughout Europe as the language of international diplomacy, the declaration notes that Haiti will not bring rebellion to other Caribbean nations, promises to respect the sovereignty of its neighbors—widely understood as a reassurance to the United States—and sets up Haiti as an example for future struggles against colonizers (an implicit reference to the many colonies then found in the Americas). So even though the declaration is explicitly addressed to the Haitian people, it's reasonable to conclude that _____

Which choice most logically completes the text?

A. aspects of the declaration were modeled on similar documents from other countries.

B. the French government may have been surprised by the declaration.

C. many Haitian people opposed the revolution and the declaration.

D. the declaration actually had several intended audiences.

ID: 09f9edb0

In the 1980s, many musicians and journalists in the English-speaking world began to draw attention to music from around the globe—such as mbaqanga from South Africa and quan họ from Vietnam—that can't be easily categorized according to British or North American popular music genres, typically referring to such music as "world music." While some scholars have welcomed this development for bringing diverse musical forms to prominence in countries where they'd previously been overlooked, musicologist Su Zheng claims that the concept of world music homogenizes highly distinct traditions by reducing them all to a single category.

Which finding about mbaqanga and quan họ, if true, would most directly support Zheng's claim?

A. Mbaqanga and quan họ developed independently of each other and have little in common musically.

B. Mbaqanga is significantly more popular in the English-speaking world than quan họ is.

C. Mbaqanga and quan họ are now performed by a diverse array of musicians with no direct connections to South Africa or Vietnam.

D. Mbaqanga and quan họ are highly distinct from British and North American popular music genres but similar to each other.

ID: 63e7799d

In vertical inheritance, parents pass genes to their offspring, but in horizontal transfer (HT), one species, often bacteria, passes genetic material to an unrelated species. In a 2022 study, herpetologist Atsushi Kurabayashi and his team investigated HT in multicellular organisms—namely, snakes and frogs in Madagascar. The team detected *BovB*—a gene transmitted vertically in snakes—in many frog species. The apparent direction of gene transfer seems counterintuitive because frogs usually don't survive encounters with snakes and so wouldn't be able to transmit the newly acquired gene to offspring, but the team concluded that *BovB* is indeed transmitted from snakes to frogs, either directly or indirectly, via HT.

Which finding, if true, would most directly support the team's conclusion?

A. *BovB* can be transmitted across frog species through HT.

B. Parasites known to feed on species of snakes and frogs in which the *BovB* gene occurs also carry *BovB*.

C. *BovB* cannot be reliably transmitted from a snake species to bacteria that are usually encountered by frog species.

D. Frog species with *BovB* show few discernible advantages as compared with frog species that do not carry *BovB*.

ID: 1a2b29c9

The following text is adapted from María Cristina Mena's 1914 short story "The Vine-Leaf."

> It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop. The doctor's hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

A. Many have come to tolerate him despite his disheveled appearance.

B. Few feel concerned that he will divulge their confidences.

C. Some dislike how freely he discusses his own family.

D. Most would be unimpressed by him were it not for his professional expertise.

ID: af9e3240

Electra is a circa 420–410 BCE play by Sophocles, translated in 1870 by R.C. Jebb. Electra, who is in mourning for her dead father and her long-absent brother, is aware of the intensity of her grief but believes it to be justified: _____

Which quotation from *Electra* most effectively illustrates the claim?

A. “O thou pure sunlight, and thou air, earth’s canopy, how often have ye heard the strains of my lament, the wild blows dealt against this bleeding breast, when dark night fails!”

B. “Send to me my brother; for I have no more the strength to bear up alone against the load of grief that weighs me down.”

C. “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied plaints, while life is in me.”

D. “But never will I cease from dirge and sore lament, while I look on the trembling rays of the bright stars, or on this light of day.”

ID: 08395130

The Hubble Space Telescope (HST) is projected to maintain operation until at least 2030, but it has already revolutionized high-resolution imaging of solar-system bodies in visible and ultraviolet (UV) light wavelengths, notwithstanding that only about 6% of the bodies imaged by the HST are within the solar system. NASA researcher Cindy L. Young and colleagues assert that a new space telescope dedicated exclusively to solar-system observations would permit an extensive survey of minor solar-system bodies and long-term UV observation to discern how solar-system bodies change over time. Young and colleagues' recommendation therefore implies that the HST _____

Which choice most logically completes the text?

A. will likely continue to be used primarily to observe objects outside the solar system.

B. will no longer be used to observe solar system objects if the telescope recommended by Young and colleagues is deployed.

C. can be modified to observe the features of solar system objects that are of interest to Young and colleagues.

D. lacks the sensors to observe the wavelengths of light needed to discern how solar system bodies change over time.

ID: d0f51067

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

A. Dog breeds would not exist without many years of human intervention in dogs' reproduction.

B. Research fails to confirm a commonly held belief about dog breeds and behavior.

C. The dog breeds most popular among owners have often changed over the past 160 years.

D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.

ID: d1539546

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh's interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____

Which choice most logically completes the text?

A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.

B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs' location.

C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.

D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

ID: 95dbdf51

Laura Mulvey has theorized that in narrative film, shots issuing from a protagonist's point of view compel viewers to identify with the character. Such identification is heightened by "invisible editing," or editing so inconspicuous that it renders cuts between shots almost unnoticeable. Conversely, Mulvey proposes that conspicuous editing or an absence of point-of-view shots would induce a more critical stance toward a protagonist. Consider, for example, the attic scene in Alfred Hitchcock's *The Birds*, a conspicuously edited sequence of tens of shots, few of which correspond to the protagonist's point of view. According to Mulvey's logic, this scene should affect viewers by _____

Which choice most logically completes the text?

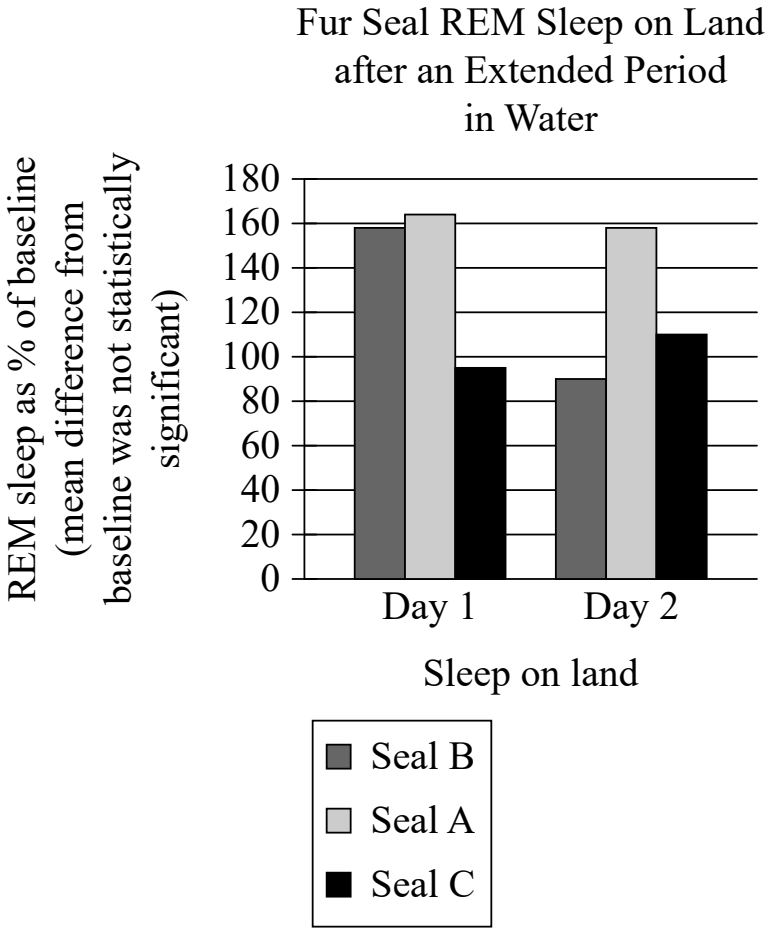
A. obscuring their awareness of the high degree of artifice involved in constructing the montage.

B. lessening their identification with the protagonist, if not alienating them from the character altogether.

C. compelling them to identify with the film's director, whose proxy is the camera, and not with the protagonist.

D. diverting their attention away from the film's content and toward its stylistic attributes.

ID: ccb1ab92



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they're in the water, they get almost none. In a study of fur seals' sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study _____

Which choice most effectively uses data from the graph to complete the text?

A. didn't show significantly less REM sleep during the second day after returning to land than they did during the first day.

B. showed no significant differences from one another in baseline levels of REM sleep.

C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.

D. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.

ID: 56f477fb

Distribution of Ecosystem Services Affected by Invasive Species by Service Type

Region (Overall)	Provisioning (75%)	Regulating (21%)	Cultural (4%)
West	73%	27%	0%
North	88%	12%	0%
South	79%	14%	7%
East	83%	6%	11%
Central	33%	67%	0%

To assess the impact of invasive species on ecosystems in Africa, Benis N. Egoh and colleagues reviewed government reports from those nations about how invasive species are undermining ecosystem services (aspects of the ecosystem on which residents depend). The services were sorted into three categories: provisioning (material resources from the ecosystem), regulating (natural processes such as cleaning the air or water), and cultural (nonmaterial benefits of ecosystems). Egoh and her team assert that countries in each region reported effects on provisioning services and that provisioning services represent the majority of the reported services.

Which choice best describes data from the table that support Egoh and colleagues' assertion?

A. Provisioning services represent 73% of the services reported for the West region and 33% of those for the Central region, but they represent 75% of the services reported overall.

B. None of the percentages shown for provisioning services are lower than 33%, and the overall percentage shown for provisioning services is 75%.

C. Provisioning services are shown for each region, while no cultural services are shown for some regions.

D. The greatest percentage shown for provisioning services is 88% for the North region, and the least shown for provisioning services is 33% for the Central region.

ID: 378c66d5

A member of the Otomi, an Indigenous people in Central Mexico, Octavio Medellín immigrated to the United States as a child, and his sculpture bears the impress of traditions on both sides of the border: US-based modernist sculpture, Mexican modernist painting, Otomi art, and the ancient sculpture of other Mexican Indigenous peoples, including the Maya. In his 1950 masterpiece *History of Mexico*, Medellín fuses these influences into a style so idiosyncratic that it resists efforts to view his work through the lens of nationality or cultural identity. Artists, he insisted, should strive for individual expression, even as they draw inspiration from their heritage and the communities where they live and work.

Which quotation from an art critic most directly challenges the underlined claim in the text?

A. "Although a number of ancient Indigenous artistic traditions pictured human forms in profile, the forms populating the surface of *A History of Mexico* suggest a specifically Maya influence."

B. "In *A History of Mexico*, the synthesis of ancient and modernist traditions functions as a stylistic parallel to the work's subject matter: a survey of centuries of Mexican history."

C. "Many critics focus on Indigenous influences in *A History of Mexico* and other key works by Medellín to the exclusion of influences from non-Indigenous art."

D. "While *A History of Mexico* features modernist motifs, it relies primarily on angular human forms in profile—a staple of Maya sculpture—and thus invites classification as Indigenous art."

ID: 0dccbf17

Henry Ossawa Tanner's 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore _____

Which choice most logically completes the text?

A. risk judging Tanner's painting by standards that may not be historically appropriate.

B. tend to conflate Tanner's political views with those of his contemporaries.

C. forgo analyzing Tanner's painting in favor of analyzing his political activity.

D. wrongly assume that Tanner's painting was intended as a critique of his fellow artists.

ID: 156ff681

Many governments that regularly transfer money to individuals—to provide supplemental incomes for senior citizens, for example—have long done so electronically, but other countries typically have distributed physical money and have only recently developed electronic transfer infrastructure. Researchers studied the introduction of an electronic transfer system in one such location and found that recipients of electronic transfers consumed a different array of foods than recipients of physical transfers of the same amount did. One potential explanation for this result is that individuals conceive of and allocate funds in physical money differently than they conceive of and allocate funds in electronic form.

Which finding from the study, if true, would most directly weaken the potential explanation?

A. Recipients of electronic transfers typically spent their funds at a slower rate than recipients of physical transfers did.

B. Nearly every recipient of an electronic transfer withdrew the entire amount in physical money shortly after receiving the transfer.

C. Recipients of physical transfers tended to purchase food about as frequently as recipients of electronic transfers did.

D. Some recipients of physical transfers received small amounts of money relatively frequently, while others received large amounts relatively infrequently.

ID: ce4448b7

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers' reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____

Which choice most logically completes the text?

A. find the disruptions more irritating as time went on.

B. rate the listening experience as more negative than those whose listening experience was uninterrupted.

C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.

D. perceive the volume of the construction noise as growing softer over time.

ID: 96802cc0

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum's long-suspected role in motor coordination.

B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.

C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.

D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

ID: 9c407117

A student performs an experiment testing her hypothesis that a slightly acidic soil environment is more beneficial for the growth of the plant *Brassica rapa parachinensis* (a vegetable commonly known as choy sum) than a neutral soil environment. She plants sixteen seeds of choy sum in a mixture of equal amounts of coffee grounds (which are highly acidic) and potting soil and another sixteen seeds in potting soil without coffee grounds as the control for the experiment. The two groups of seeds were exposed to the same growing conditions and monitored for three weeks.

Which finding, if true, would most directly weaken the student's hypothesis?

A. The choy sum planted in the soil without coffee grounds were significantly taller at the end of the experiment than the choy sum planted in the mixture of soil and coffee grounds.

B. The choy sum grown in the soil without coffee grounds weighed significantly less at the end of the experiment than the choy sum grown in the mixture of soil and coffee grounds.

C. The choy sum seeds planted in the soil without coffee grounds sprouted significantly later in the experiment than did the seeds planted in the mixture of soil and coffee grounds.

D. Significantly fewer of the choy sum seeds planted in the soil without coffee grounds sprouted plants than did the seeds planted in the mixture of soil and coffee grounds.

ID: f9bd4e61

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht's influence can be seen in English playwright Caryl Churchill's 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____

Which choice most logically completes the text?

A. focus on the characters' beliefs about social and political issues as revealed by the characters' actions.

B. reflect on social and political phenomena not directly related to patriarchy and colonialism.

C. recognize pertinent social and political parallels between Germany during Brecht's time and England at the time when Churchill was writing *Cloud 9*.

D. be dispassionate as they think critically about the social and political questions raised by the play.

ID: dd1757fd

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.

Which finding, if true, would most directly support the claim in the underlined sentence?

A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.

B. The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.

C. Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.

D. Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.

ID: a44bbd6b

Several studies of sediment (e.g., dirt, pieces of rock, etc.) in streams have shown an inverse correlation between sediment grain size and downstream distance from the primary sediment source, suggesting that stream length has a sorting effect on sediment. In a study of sediment sampled at more than a dozen sites in Alpine streams, however, geologists Camille Litty and Fritz Schlunegger found that cross-site variations in grain size were not associated with differences in downstream distance, though they did not conclude that downstream distance is irrelevant to grain size. Rather, they concluded that sediment influx in these streams may have been sufficiently spatially diffuse to prevent the typical sorting effect from being observed.

Which finding about the streams in the study, if true, would most directly support Litty and Schlunegger's conclusion?

A. The streams regularly experience portions of their banks collapsing into the water at multiple points upstream of the sampling sites.

B. The streams contain several types of sediment that are not typically found in streams where the sorting effect has been demonstrated.

C. The streams mostly originate from the same source, but their lengths vary considerably due to the different courses they take.

D. The streams are fed by multiple tributaries that carry significant volumes of sediment and that enter the streams downstream of the sampling sites.

ID: 80fd9454

Percentage of Available Eggs Eaten by Cane Toad Tadpoles

Amphibian species (common name)	Percentage of eggs eaten	Native to Australia	Produces bufadienolide
Little red tree frog	1%	yes	no
Cane toad	90%	no	yes
Short-footed frog	7%	yes	no
Striped burrowing frog	10%	yes	no
Dainty green tree frog	1%	yes	no

Native to Latin America, the cane toad was introduced to Australia in the 1930s. In recent decades, tadpoles in the Australian population have been shown to consume eggs of their own species. A 2022 study showed that when presented with cane toad eggs as well as eggs of native Australian amphibians, cane toad tadpoles disproportionately consumed eggs of their own species. This behavior results from their attraction to bufadienolide, a chemical produced by the eggs of cane toads but not by the eggs of native amphibians. However, using data from this study, a student wishes to argue that the presence of bufadienolide doesn't entirely explain the cane toad tadpoles' preference for certain eggs over others.

Which choice best describes data from the table that support the student's argument?

A. The tadpoles consumed a higher percentage of the striped burrowing frog eggs than they did of the eggs of the dainty green tree frog.

B. The tadpoles left a certain percentage of the eggs of each of the five species unharmed, thus ultimately allowing them to hatch.

C. The tadpoles consumed a lower percentage of the short-footed frog eggs than they did of the eggs of their own species.

D. The tadpoles consumed the same percentage of the dainty green tree frog eggs as they did of the little red tree frog eggs.